

**Ch-22 Accident Theories &
Investigation
Homework Answers**



01

The best practices drivers should be trained to complete after an accident (when safe to do so) include:

- A. Photographing involved vehicle(s) from all sides.**
- B. Collect photographs of injured person(s).
- C. Detailed drawings of skid marks and distances.
- D. Photographs of sky and weather-related details.



02

Two events in a fault tree have probabilities of 0.80 and 0.20, respectively. The events are joined by an OR gate. What is the numerical probability of the TOP event associated directly with this OR gate?

A. 0.12

B. 0.16

C. 0.80

D. 0.84

Rare Events Correlation, $(P_a + P_b) - (P_a \times P_b) = (0.8+0.2) - (0.8 \times 0.2) = 0.84$



03

What are the five dominoes in Heinrich's Domino Theory of incident causation?

- A. Injury, incident, unsafe act, unsafe condition, undesirable trait
- B. Injury, incident, unsafe act, uncontrolled energy, social environment
- C. Injury, incident, unsafe condition, uncontrolled energy, social environment
- D. Injury, incident, unsafe act or unsafe condition, undesirable traits, social environment**



04

The primary reason for conducting an accident investigation is to:

- A. Reduce the cost for Workers' Compensation
- B. Find and fix the blame
- C. Identify the cause and take actions to prevent recurrence**
- D. Prepare reports needed for litigation



05

In fault tree analysis, two types of logic gates are used. Which statement below is correct?

- A. The either/or gate is the most common.
- B. An AND gate means that all contributing events connected to the main event must occur in order for the main event to occur.**
- C. An OR gate means that all contributing events connected to the main event must occur in order for the main event to occur.
- D. An AND gate means that if either of the contributing events connected to the main gate occur, the main event will also occur.



06

What is characteristic of a root cause of an incident?

- A. An intentional act made by an employee
- B. An error made by an employee
- C. An error made by a manager
- D. A management system problem**



07

Which of the following is NOT an example of a root cause of an accident??

- A. A worker made a mistake due to a lack of training.
- B. A machine malfunctioned due to poor maintenance.
- C. A company did not provide adequate safety equipment.
- D. An employee had a pre-existing medical condition.**



08

Engineers are designing a new rocket to carry people to the moon. After the initial design is complete, they will evaluate each component to determine what is likely to happen in the event the component fails. What is this type of analysis called?

- A. Hazard and Operability Study (HAZOP)
- B. Failure Modes and Effect Analysis (FMEA)**
- C. What-If Analysis
- D. Fault Tree Analysis



09

Which of the following system safety analysis techniques may lead to an estimation of the probability of a catastrophic event?

A. Management Oversight and Risk Tree

B. Fault Tree Analysis

C. Preliminary Hazard Analysis

D. Failure Modes and Effect Analysis

A fault tree analysis performed for a small system identified five basic initiating events having probabilities of occurrence as shown here in the following table.

The minimal cut sets from the fault tree analysis are shown below.

- AB
- CD
- DE

What is the TOP event probability for this fault tree?

Event	Probability
A	10^{-5}
B	10^{-5}
C	10^{-7}
D	10^{-3}
E	10^{-6}

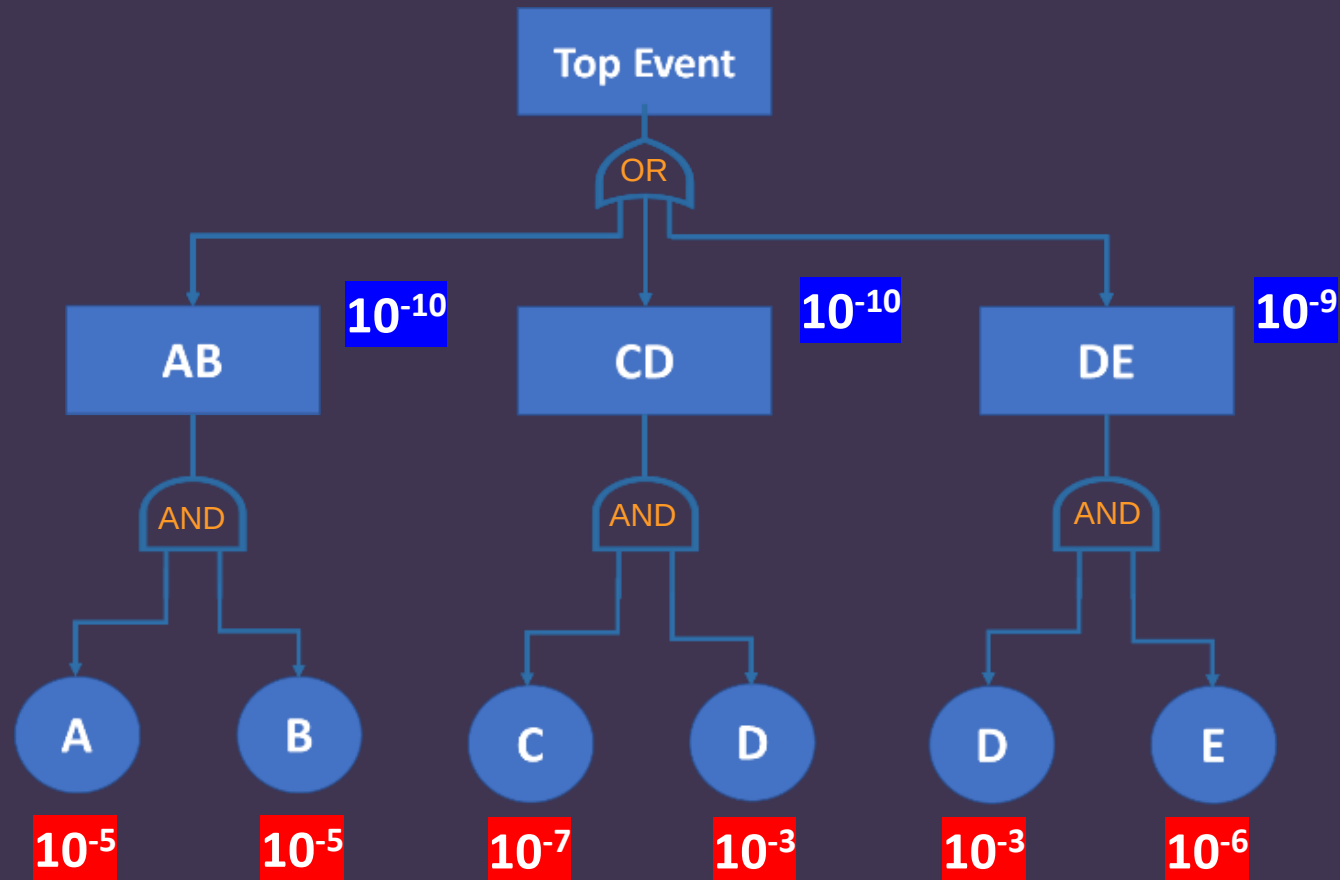
10

A. 4.0×10^{-9}

B. 1.2×10^{-9}

C. 3.3×10^{-26}

D. 3.3×10^{-29}



$$\text{TOP Event Probability} = (10^{-5} \times 10^{-5}) + (10^{-7} \times 10^{-3}) + (10^{-3} \times 10^{-6}) = 1.2 \times 10^{-9}$$